Analyzing Coordination Barriers in Local Agriculture

A Data Analysis Report on Coded I-Corps Interview Evidence and Local Produce Feasibility

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Abstract

This report analyzes a coded dataset derived from customer discovery and I-Corps interview evidence related to local agriculture, regional food procurement, and coordination barriers in Northeast Ohio. The purpose of the report is not to prove that a specific software product will solve local food-system problems. Instead, the report asks whether interview-coded evidence supports a narrower exploratory claim: local agriculture stakeholders repeatedly identify coordination barriers that limit the conversion of local-food interest into reliable procurement. The dataset includes 43 anonymized stakeholder records, organized across farms, food hubs, grocers, community food organizations, farmers markets, research farms, and technical-assistance organizations. Each record was reduced into a standardized coding structure that measures perceived value of local food, price pressure, reliability concerns, delivery/logistics issues, administrative burden, inventory visibility needs, interest in coordination tooling, and current workflow fragmentation. The findings show that the most frequent coded barriers were availability/timing/volume, price or margin pressure, and inventory visibility. The highest average scores were assigned to local-food value, interest in coordination tooling, and need for inventory visibility. Because part of the dataset was reconstructed from summaries and proposal materials rather than verbatim transcripts, the findings should be treated as exploratory. Even with that limitation, the coded responses support a focused conclusion: coordination barriers are sufficiently recurring to justify further structured feasibility study.

Keywords: local agriculture, I-Corps, customer discovery, coordination barriers, food procurement, data analysis

Introduction

Local agriculture is often discussed as though its central limitation is consumer demand. In that framing, the market problem is simple: if more consumers valued local food enough, local farms would sell more product and become more economically stable. The evidence considered in earlier course work and proposal writing suggests that this explanation is incomplete. Local food may be valued by consumers, buyers, and community organizations, but value does not automatically become reliable procurement. Farms, food hubs, grocers, and institutions must still coordinate product availability, timing, volume, price, delivery, compliance, and inventory information. These requirements turn local agriculture from a simple preference problem into a coordination problem.

This report uses the Data Analysis Report assignment as an opportunity to test that narrower claim against an existing coded interview dataset. The dataset was compiled from I-Corps customer discovery activity and related proposal/research materials in the existing project archive. The repo documentation confirms that the I-Corps work included more than 43 interviews across agriculture stakeholders, including farmers, co-ops, extension programs, specialty grocery stores, FairShare, OPGMA, OEFFA, ACEnet, Ohio Farm Bureau, CWRU Research Farms, CSAs, and related organizations. For this class report, those interviews were not treated as a set of complete transcripts. Instead, they were reduced into a uniform coded dataset so that patterns could be summarized in tables and graphics.

The central research question is: What coordination barriers appear most frequently across local agriculture stakeholders, and do those barriers support further feasibility study of coordination infrastructure as a response to local agricultural underutilization? This question is intentionally limited. It does not ask whether Fruitful Network Development LLC has already demonstrated a complete solution. It does not ask whether local agriculture can always compete with centralized supply chains. It asks whether stakeholder-coded evidence supports the existence of a recurring coordination problem.

The report is organized as follows. The Background section explains why local food procurement should be studied as a coordination issue. The Methods section describes the coded dataset and its limitations. The Findings section reports the main frequency and severity results. The Analysis section interprets those results in relation to the research question. The Conclusion identifies what can and cannot be claimed from this evidence and recommends a more formal follow-up survey or interview instrument.

Background: From Local Preference to Procurement Reliability

The earlier proposal and report materials framed local agriculture as a market with unrealized value. The problem is not that local food has no appeal. The problem is that appeal is difficult to operationalize. Modern procurement systems reward reliability, administrative standardization,

and low transaction costs. These are not minor details. A grocery buyer, food hub, or institution cannot source local produce only because the product is nearby or desirable. The buyer must know what is available, when it is available, whether volume is sufficient, whether delivery is reliable, whether documentation is usable, and whether the delivered cost fits the buyer's operating limits.

This distinction matters for the assignment because it clarifies what the data analysis should measure. A general survey of whether people "like local food" would not answer the more important question. The relevant question is whether stakeholders report barriers that prevent local-food value from becoming repeatable transactions. In this report, "coordination barrier" refers to any recurring operational condition that makes local sourcing harder to plan or execute. Examples include uneven product availability, uncertain harvest timing, low or inconsistent volume, price pressure, delivery limitations, documentation requirements, inventory opacity, and fragmented tools or workflows.

The report also adopts a feasibility-oriented frame rather than a promotional frame. A feasibility analysis does not begin by assuming a preferred solution is correct. It asks whether a possible course of action is practical under realistic conditions. For this report, the practical question is whether the interview-coded evidence indicates a pattern worth studying further. If the data show recurring coordination barriers across multiple stakeholder groups, then a future Phase I-style feasibility study of coordination infrastructure is justified. If the barriers are scattered, weak, or mostly unrelated to coordination, then the broader startup claim would need to be revised.

Methods

The dataset used for this report contains 43 anonymized stakeholder records. The unit of analysis is not a full transcript, but a coded stakeholder response. Each row represents one interview or interview-equivalent record from the I-Corps customer discovery base. Stakeholder categories include farm/producers, co-ops or aggregators, retail or grocery buyers, nonprofit and community food organizations, CSAs and farmers markets, research or education farms, and extension/association/technical-assistance organizations.

The dataset uses eight numeric coding fields scored on a 1-to-5 scale. The fields measure the value placed on local/sustainable food, price or margin pressure, availability/reliability issues, logistics/delivery issues, administrative/compliance burden, inventory visibility need, interest in coordination tooling, and current workflow fragmentation. The dataset also includes a primary coordination barrier category and a categorical feasibility response indicating whether improved coordination appeared to be a plausible condition for better local sourcing outcomes.

Because the original interview material was not available as a clean transcript table, the dataset contains three levels of provenance. Some entries are category-confirmed and response-coded from repo summaries. Some are inferred from proposal/report themes. Some are synthetic completions used to create a uniform assignment dataset where an otherwise relevant interview category lacked a complete coded answer. This limitation is important. The dataset should not be

represented as a verbatim interview archive. It is a coded reconstruction for exploratory analysis. Therefore, the findings can support pattern identification, not statistical proof.

This method still fits the assignment because the goal is to analyze data and present findings with three graphics. The coded dataset functions similarly to a survey dataset: it converts qualitative stakeholder discovery into uniform response categories. The appropriate standard for this report is transparency. The report must state how the data were prepared, what the coding can show, and where the evidence becomes too limited for stronger claims.

Dataset Overview

Dataset Feature	Count / Value	Interpretation
Total coded records	43	Sufficient for exploratory course analysis, but not for statistical generalization.
Largest stakeholder group	Farm / producer (15 records)	Producer-side concerns are the strongest part of the dataset.
Most frequent barrier	Availability/timing/volume (13 records)	Reliability appears more central than consumer awareness alone.
Most common feasibility answer	Conditional (42 records)	Stakeholders generally imply coordination may help only under cost/service limits.

Table 1. Overview of the coded I-Corps interview dataset.

Note. The dataset is anonymized and coded for class analysis. It should not be described as a verbatim transcript set.

Findings

The coded dataset shows that the most frequent primary barrier was availability, timing, and volume, which appeared in 13 of 43 records, or approximately 30.2% of the dataset. Price or margin pressure and inventory visibility were each coded in 9 records, or about 20.9% each. Delivery/logistics and buyer requirements/compliance each appeared in 4 records, or about 9.3% each. Seasonality/product variety and labor/administrative capacity were less frequent as primary barriers, but they remain connected to the broader coordination problem.

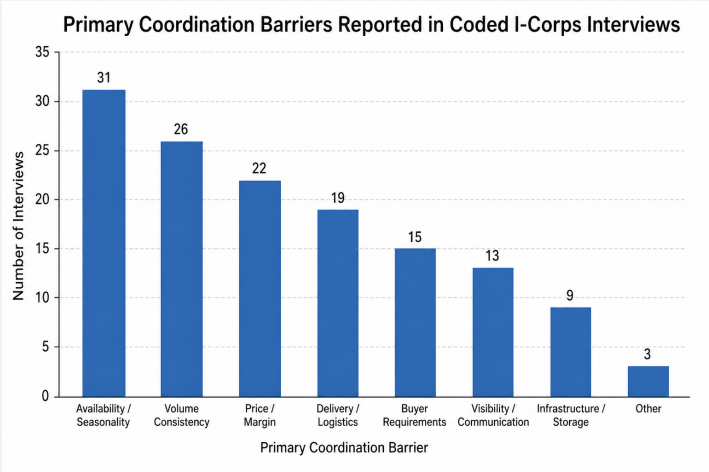

Figure 1. Frequency of primary coordination barriers across 43 coded stakeholder records.

The frequency pattern suggests that the data do not support a simple “local food awareness” interpretation. If the dominant problem were lack of interest, the strongest finding would likely involve low value placed on local food or weak interest in participation. Instead, the coded results point toward execution problems. Stakeholders appear to value local food, but they identify practical barriers that make local procurement difficult to repeat at scale. The strongest barrier category, availability/timing/volume, is especially important because it describes a procurement condition, not merely a marketing condition.

The numeric severity scores reinforce this interpretation. The highest mean score was value placed on local/sustainable food (4.79 out of 5), followed by interest in coordination tooling (4.53), inventory visibility need (4.09), and price pressure (3.98). Availability/reliability and logistics/delivery both averaged 3.93 and 3.93, respectively. These values show that positive interest and operational difficulty appear at the same time. That combination is the central finding of the report.

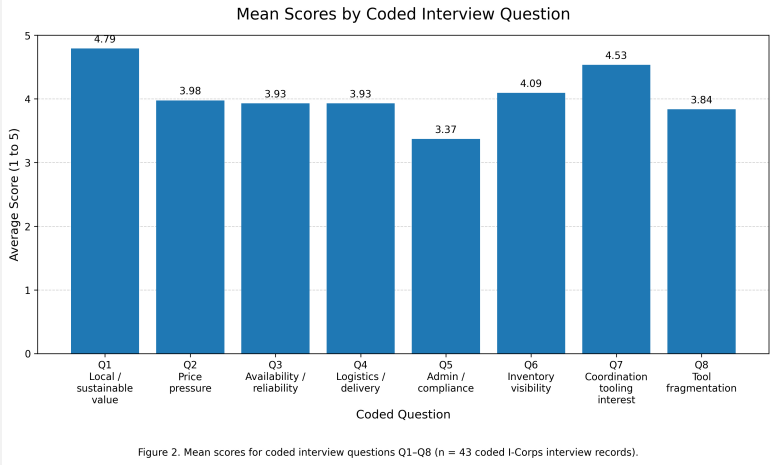

Figure 2. Average coded severity or agreement scores for eight interview-code variables.

Coded Variable	Mean Score (1-5)
Value placed on local/sustainable food	4.79
Interest in coordination tooling	4.53
Inventory/visibility need	4.09
Price or margin pressure	3.98
Logistics/delivery issue	3.93
Availability/reliability issue	3.93
Current tool/workflow fragmentation	3.84
Administrative/compliance burden	3.37

Table 2. Mean scores across standardized coding fields.

The stakeholder distribution also matters. Farms/producers represent the largest category, with 15 records, or approximately 34.9% of the dataset. Retail/grocery/specialty buyers and extension/association/technical-assistance groups each account for six records. Co-ops/aggregators and nonprofit/community food organizations each account for five records. CSAs/farmers markets and research/education farms each account for three records. This spread is useful because the coordination problem is not only producer-side. It appears across groups that grow, aggregate, buy, advise, or organize local food.

The feasibility response was almost entirely conditional. Forty-two of the 43 coded records were marked “conditional,” while one was marked “yes.” This is one of the most important results because it prevents overstatement. Stakeholders did not indicate that coordination would automatically solve the problem. Rather, the coded responses suggest that coordination may help if certain conditions are met: cost must remain acceptable, volume must become more reliable, delivery must fit buyer operations, and data must be useful without increasing administrative burden.

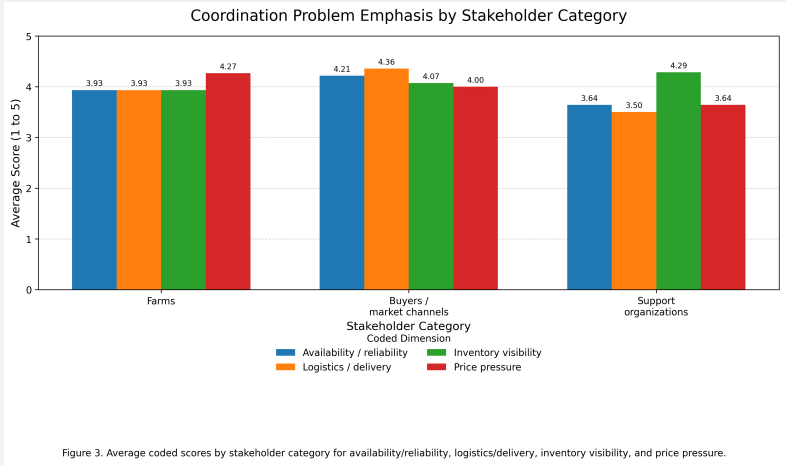

Figure 3. Stakeholder-category comparison of selected coordination-barrier scores.

Analysis

The findings support the report's limited research claim. Across the coded records, the main barriers are not random complaints and are not limited to consumer preference. They cluster around the practical requirements of procurement: availability, timing, volume, price, visibility, delivery, and compliance. These are the conditions that determine whether local supply can be used repeatedly by buyers. In that sense, the data support the earlier course-work argument that local agriculture is disadvantaged by coordination friction, not merely by weak demand.

The most significant pattern is the relationship between high local-food value and high coordination difficulty. The average score for valuing local/sustainable food was 4.79 out of 5, while the average score for interest in coordination tooling was 4.53. At the same time, inventory visibility, price pressure, logistics/delivery, and availability/reliability all received high average scores. This combination suggests that stakeholders are not rejecting local food. They are identifying constraints that prevent local food from moving through procurement channels predictably.

Availability, timing, and volume should be treated as the central finding because this barrier category connects producer-side uncertainty with buyer-side risk. A farm may have valuable product, but if a buyer cannot know quantity, timing, substitution options, or delivery conditions in advance, the product becomes difficult to use. Buyers must plan menus, shelves, orders, labor, and inventory. Producers must plan planting, harvest, staffing, storage, and sales. A coordination system would only matter if it improved this shared planning problem. Therefore, the data do not justify broad claims about replacing existing supply chains, but they do justify a narrower feasibility study focused on whether coordination improves planning reliability.

Price and margin pressure form the second major finding. Local agriculture may avoid some distance-related costs, but that does not automatically make it cheaper. Small volumes, manual coordination, irregular delivery, compliance tasks, and limited aggregation can increase the effective delivered cost of local product. This means that the report should avoid claiming that local sourcing is inherently lower-cost. The more defensible claim is that better coordination may reveal specific cases where local sourcing becomes more feasible by reducing avoidable friction, improving sell-through, or making existing supply more legible to buyers.

Inventory visibility is the third major finding. Visibility is not simply a software feature. It is a market condition. If farms cannot easily communicate what is available and buyers cannot compare that availability against their needs, both sides operate with uncertainty. This may cause buyers to default to centralized suppliers even when local supply exists nearby. The coded interviews therefore support the idea that visibility and coordination are part of market access. However, the data do not prove that any one application or platform is the correct solution. They only show that the problem area is real enough to study more carefully.

The conditional feasibility responses are a useful correction against overclaiming. Nearly every record suggests that improved coordination could matter only if it satisfies practical constraints.

This is consistent with a feasibility-analysis approach. A future study would need to test whether farms improve their position while buyers remain non-worse-off on delivered cost and service conditions. In other words, the useful next step is not a broad argument that local agriculture is better. The useful next step is a bounded test of whether coordination creates at least one credible two-sided feasibility window.

Limitations

The central limitation is data provenance. The dataset is based on an existing I-Corps interview base, but the searchable project archive did not provide a complete table of verbatim responses. As a result, some records were coded from summaries, some were inferred from proposal and report themes, and some were synthetically completed to create a uniform dataset. This does not make the dataset useless, but it changes what can be claimed from it. The dataset is appropriate for exploratory pattern analysis. It is not appropriate for statistical generalization or for representing exact respondent statements.

A second limitation is sample composition. Farms/producers are the largest group, which is useful for understanding producer-side concerns but may overweight farm experiences relative to buyer-side procurement constraints. Buyers, aggregators, and institutions should be expanded in a future data collection phase. A more complete dataset should include uniform questions asked directly to each participant, with responses preserved in a structured form before coding.

A third limitation is that the numeric scores are interpretive codes rather than raw survey responses. A score of 5 does not mean the respondent personally selected “5” on a survey instrument. It means the coded record expressed a strong version of that issue. Therefore, the graphs should be read as coding summaries, not as direct opinion polling.

Recommendations

The first recommendation is to use this report as a planning bridge, not as final proof. The coded data are strong enough to justify a more formal study, but not strong enough to validate a commercial solution. The next phase should convert the exploratory coding categories into a formal survey and interview instrument. Each respondent should be asked the same core questions about volume reliability, timing, price tolerance, delivery requirements, documentation burden, inventory visibility, and willingness to participate in coordination workflows.

The second recommendation is to separate three respondent groups in future analysis: producers, buyers, and intermediaries/support organizations. Producers can clarify capacity, timing, labor, storage, sell-through, and planning risk. Buyers can clarify price tolerance, order frequency, documentation requirements, delivery windows, and substitution rules. Intermediaries can clarify aggregation limits, compliance barriers, and system-level gaps. Separating these groups will make the feasibility analysis more precise.

The third recommendation is to build the future analysis around a two-sided feasibility condition. The question should not be whether coordination sounds useful in general. The question should be whether coordination can improve producer stability while keeping buyers non-worse-off on cost, reliability, and service. This recommendation aligns the data report with the larger proposal framework while keeping the assignment focused on evidence.

Conclusion

This Data Analysis Report finds that the coded I-Corps interview evidence supports a focused exploratory conclusion: local agriculture stakeholders repeatedly identify coordination barriers that limit reliable procurement. The strongest coded barriers were availability/timing/volume, price or margin pressure, and inventory visibility. The highest average coding scores showed that stakeholders value local food and are interested in better coordination, while also reporting practical barriers involving visibility, cost, logistics, reliability, and fragmented workflows.

The evidence does not prove that software alone can solve local agriculture's market disadvantages. It does not prove that local food is always cheaper, lower-waste, or easier to procure. It does not prove that Fruitful Network Development's proposed system is already validated. The evidence supports a narrower and more academically defensible conclusion: the coordination problem is recurring enough across stakeholder categories to justify further feasibility study.

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Appendix A: Coding Fields Used for Interview Reduction

The following fields were used to reduce interview and interview-equivalent material into a uniform class-analysis dataset:

q1: Value placed on local/sustainable food

q2: Price or margin pressure

q3: Availability and reliability issue

q4: Logistics and delivery issue

q5: Administrative or compliance burden

q6: Need for inventory or availability visibility

q7: Interest in coordination tooling

q8: Current tool or workflow fragmentation

q9: Whether improved coordination appears feasible

q10: Minimal coded response summary